

# TrES-3b

R-2616

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_TRES-3\_260711 · created 2026-08-28T22:53:55+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

## Results

|                                                   |                                |
|---------------------------------------------------|--------------------------------|
| Mid-Transit Time (Tmid)                           | 2461232.7844 +/- 0.002 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.187 +/- 0.057                |
| Transit depth (Rp/Rs)^2                           | 3.49 +/- 2.13 [%]              |
| Orbital Inclination (inc)                         | 82.29 +/- 2.95                 |
| Airmass coefficient 1 (a1)                        | 0.91 +/- 0.58                  |
| Airmass coefficient 2 (a2)                        | 0.34 +/- 1.46                  |
| Scatter in the residuals of the lightcurve fit is | 66.02 %                        |
| Best Comparison Star                              | #3 - [348, 68]                 |
| Optimal Aperture                                  | 0.75                           |
| Optimal Annulus                                   | 3.75                           |
| Transit Duration (day)                            | 0.052 +/- 0.027                |

## Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                            |
|---------------------------|--------------------------------------------|
| Rp/R■ fit vs published    | 0.187 ± 0.057 vs 0.16309 (deviation 0.42σ) |
| Duration fit vs published | 0.052 d vs 0.0567 d (deviation 0.17σ)      |
| Mid-time O–C              | -3.4 min                                   |
| Depth SNR                 | 3.3                                        |
| Residual scatter          | 66.02 %                                    |

## Light curve

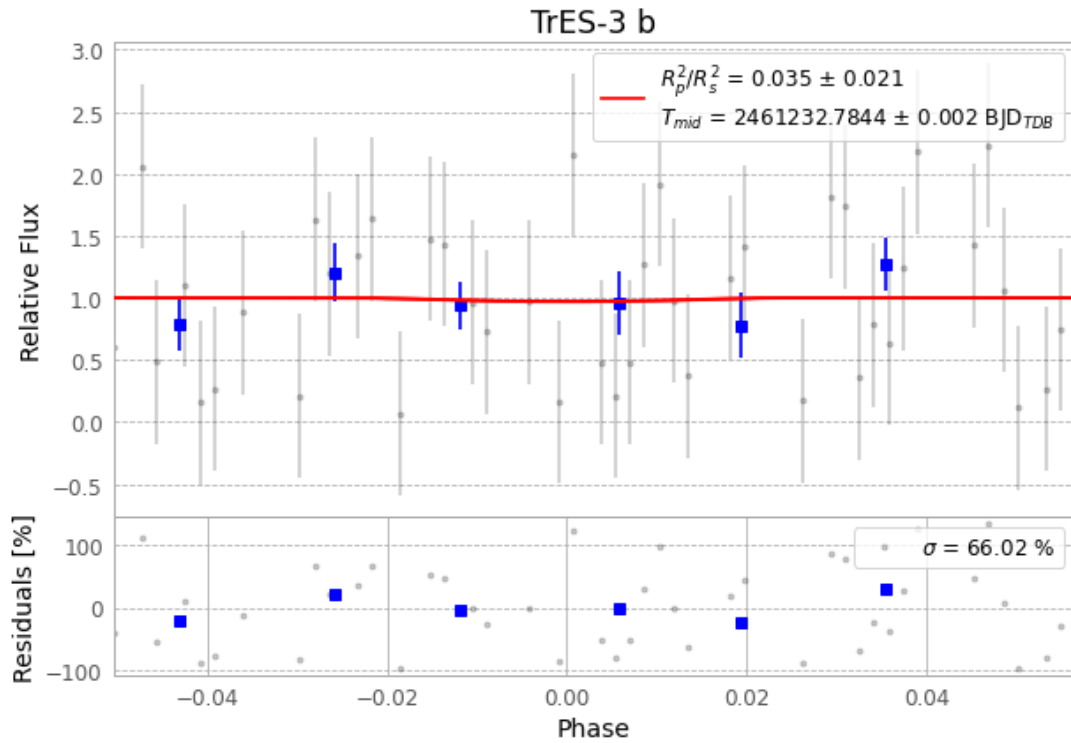


Figure 1 — FinalLightCurve\_TrES-3 b\_2026-07-10.png (SHA-256 e6a393fa4617005e...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [182, 300], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 1a3301df8f5cb8cc...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit a4253ce

## Data provenance

68 FITS frames (45 MB), TRES-3260711050915.FITS → TRES-3260711083016.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-260711.log (649aac9cd409...), FinalLightCurve\_TrES-3 b\_2026-07-10.png (e6a393fa4617...), FinalLightCurve\_TrES-3 b\_2026-07-10.csv (e7123d71eae6...)

## Compute resources

Wall time 118.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-28 22:53Z.